P-Workshop - 27 Feb 2026

Phonologically conditioned allomorphy in Mapudungun

A missed generalisation or a historical accident?

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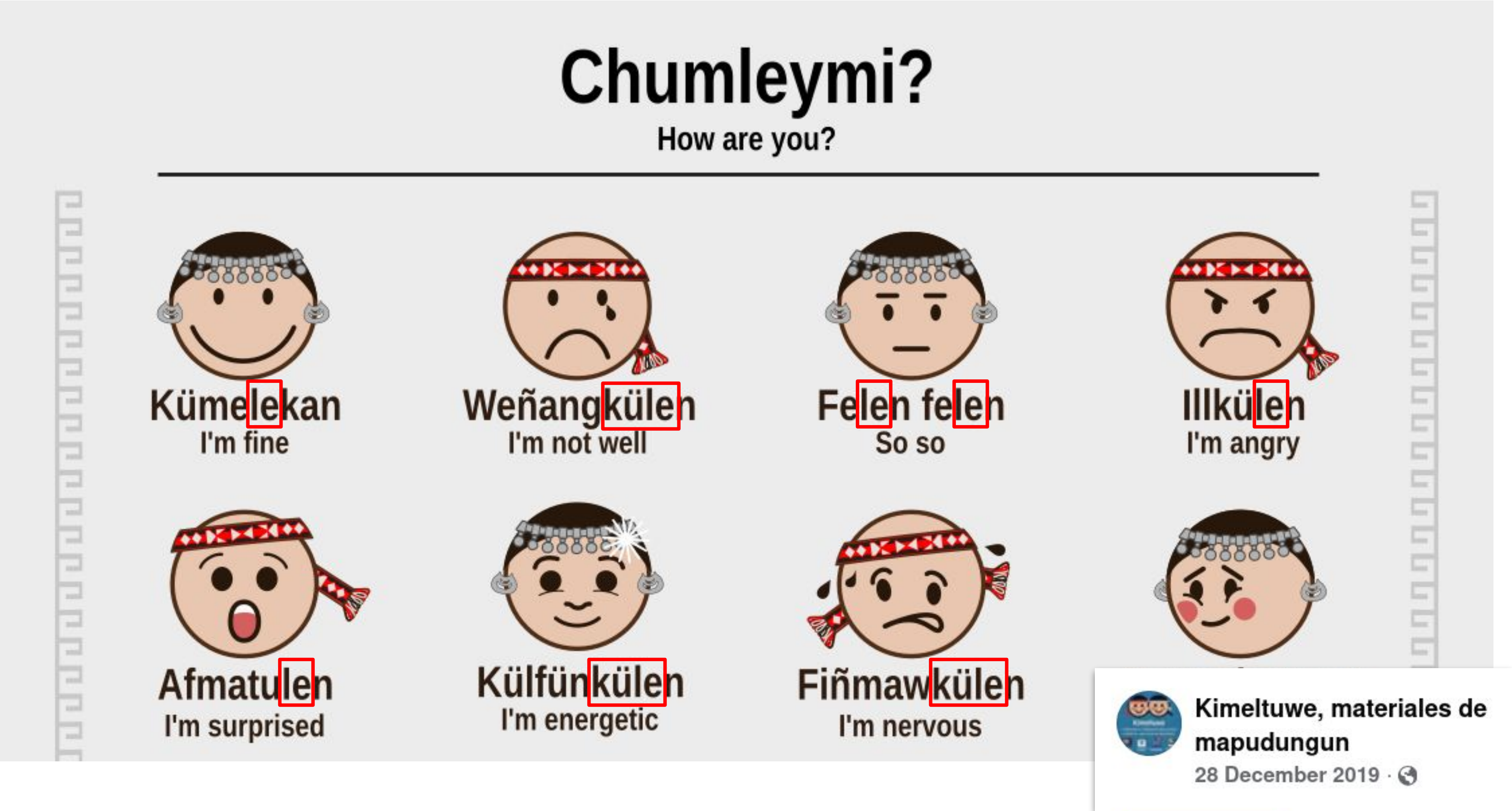
University of Edinburgh



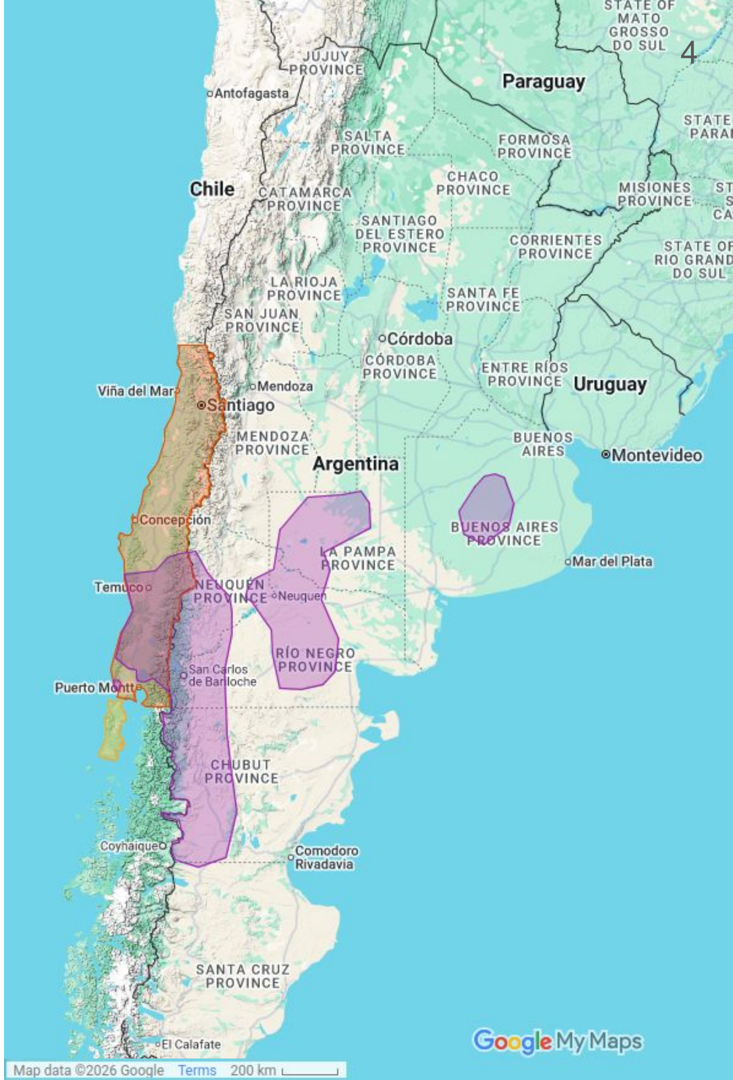
Kimeltuwe, 2019

My assertions for this P-Workshop talk

- **Kɨ-allomorphy is:**
  - Phonologically-conditioned allomorphy (PCA)
  - Suppletive allomorphy (PCSA)
  - Input-Subcat: Despite appearances, it is not phonologically optimising.
- **Why is this important?**
  - To elucidate the division of labour between computation and storage
  - To explore how this phenomenon behaves in a language with complex morphology and separative exponence

Layout

- About Mapudungun
- Corpus-based survey
- PCA typology
- What is kɨ-allomorphy?
- Conclusions



Areas where mapudungun was historically spoken:
1600s in yellow; 1900s in purple (Adelaar 2004)

Mapudungun phonemic inventory and phonotactics

(C)V(C)

ONSET

p	t̪	t	ʈʂ	tʃ	k
f	θ	s	ʐ	ʃ	ɣ
m	n̪	n		ɲ	ŋ
w	l̪	l		ʎ,j	

NUCLEUS

i	ɨ	u
e		o
	a	

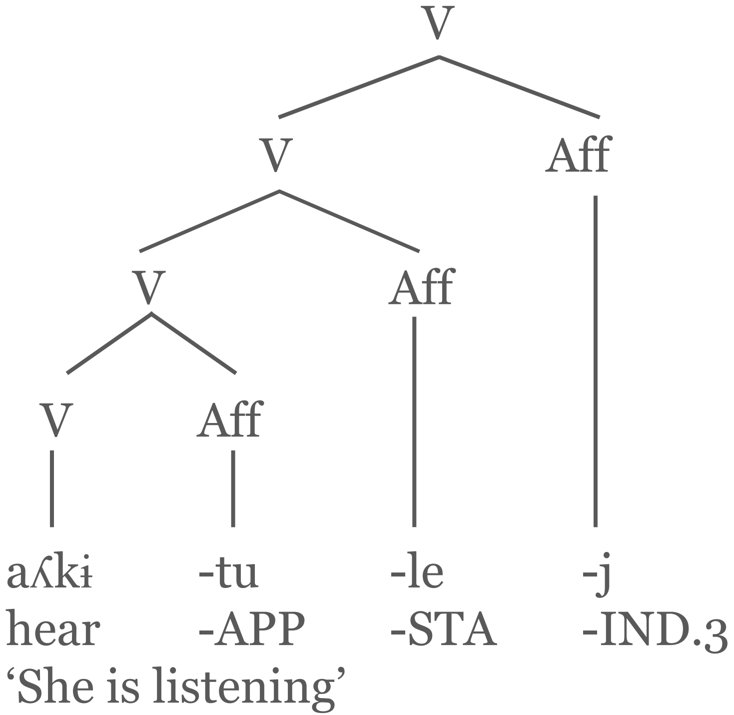
CODA

—	—	—	—	—	—
f	θ	s	ʐ	ʃ	ɣ
m	n̪	n		ɲ	ŋ
w	l̪	l		ʎ,j	

Sadowsky et al. 2013

Morphology

- Separative exponence / suffixing
- Bare and complex bases
  - Derivational suffixes
  - Valency-changing suffixes
- Compounds
  - Serial verbs
  - Noun incorporation
  - Reduplication
- Obligatory inflection
  - mood, person, number
- Other inflectional suffixes
  - Aspect, tense, evidentiality, space-related values, etc.



Smeets 2008, Zúñiga 2015, Molineaux 2024

Kɨ-allomorphy

Kɨ-allomorphs I

Inflection affixes with two allomorphs after a morpheme ending in a vowel (V) or a consonant (C):

(1) Stative C–kɨle vs. V–le "to be doing"

	After C	meaning
a.	kon-kɨle-	'be entering'
b.	lif-kɨle-	'be clean'
c.	umaw-kɨle-	'be sleeping'
d.	pot͡ʂoŋ-kɨle-	'be bent down'
e.	ɲɨkɨf-kɨle-	'be silent'

	After V	meaning
f.	pi-le-	'be saying'
g.	aʎfɨ-le-	'be injured'
h.	afmatu-le-	'be surprised'
i.	longko-le-	'be acting as leader'
j.	moŋe-le-	'be living'

Küme Dungu Corpus (Sandvig, 1985)

Kɨ-allomorphs II

(2) Ambulative C-kɨjaw ~ V-jaw 'to go round doing'

	After C	meaning
a.	ɲam-**kɨjaw**-	'be lost; be wandering aimlessly'
b.	lef-**kɨjaw**-	'be running around'
c.	kɨθaw-**kɨjaw**-	'be working, to be busy working'

	After V	meaning
a.	t͡ʂeka-**jaw**-	'be walking around'
b.	mɨpɨ-**jaw**-	'be flying around'
c.	awkantu-**jaw**-	'to be playing around'

Küme Dungu Corpus (Sandvig, 1985)

Corpus-based survey

Survey of the stative and ambulative markers

- 21 books written in Mapudungun between 1987 and 1998
- Number of words: 62587
- Total kɨ-allomorphy: 1369
    - Stative: 1244 Ambulative: 125

Mª Luisa Cayuman, José B. Ancan, Eleuterio Cayulao and Antonio Canio. Proyecto Wiñokintun ñi wirin (Caniguan et al. 2024)


Relative dispersion
PND (188/6560)
CNKN (91/4777)
NCH (91/4566)
KTN (73/4002)
FED2 (45/3817)
FED1 (82/3776)
FED4 (68/3662)
FED3 (77/3622)
CNN (42/3577)
TPU (62/3536)
DNK (67/3295)
KTD2 (74/2548)
PTN (57/2425)
KWT (49/2377)
CTM2 (52/1978)
FTM1 (22/1758)
PDG (24/1722)
PMK (32/1691)
FTM2 (14/1253)
LPTC (17/950)
CTM1 (4/695)
0%
25%
50%
75%
100%
Token position in document (relative)
Suffix
küle
le

Summary results of the corpus-based exploration

- Kɨ-allomorphy is widespread and productive.
- No morphological or lexical conditioning: the forms occur after several types of bases:
  - Native forms and loanwords
  - Bare stem and derived stems
  - Compounds
- The conditioning factor is phonological (last C / V), and it is not affected by the number of syllables.

But what type of PCA is kɨ-allomorphy?

Phonologically conditioned Allomorphy (PCA)

—a typology

What is allomorphy?

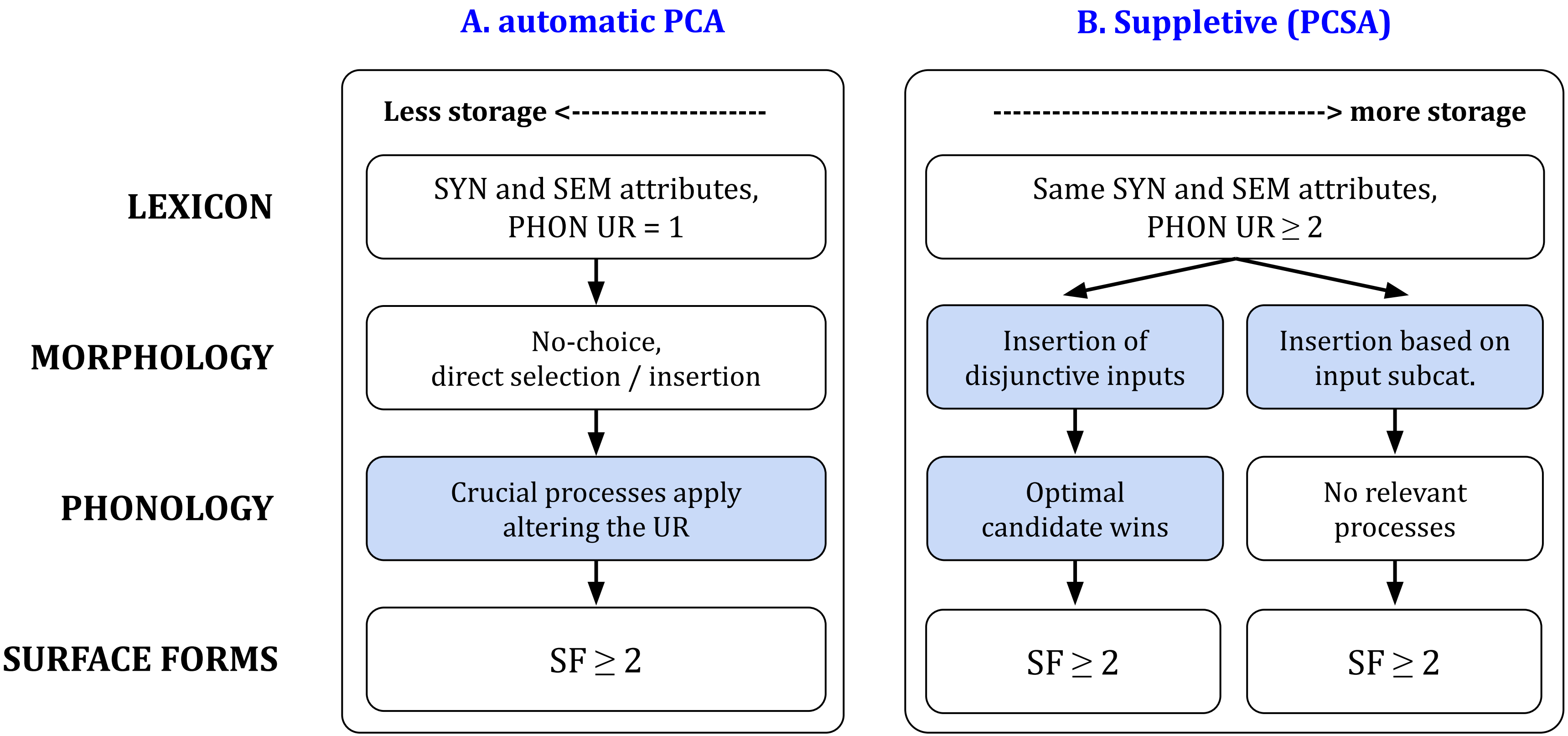
- **Allomorphy** is the relationship between different surface forms that have the same meaning.
- The conditioning factors can be **phonological**, morphological or lexical.
- It can target the phonology of stems and affixes

Why PCA phenomena are important?

- Phonologically conditioned alternations are relevant for a fundamental question in the conceptualisation of grammar: **Where is the separation between storage and computation?** (Mascaró 1996, Nevins 2011, Bermúdez-Otero 2012, Embick and Shwayder 2018)

Simplified model of PCA allomorphy

Kiparsky 1996, Bermúdez-Otero 2012, 2013, Paster 2014, Armstrong 2021, Bonet 2023



Types of PC allomorphy

- **PC Allomorphy**
  - **a. Automatic**
  - **B. Suppletive**
    - **b1. Optimising**
    - **b2. Arbitrary**

In the examples:
Allomorphs are shown in **red**,
the phonological condition in **blue**

BASE-TRIGGER-ALLO$_1$

BASE-TRIGGER-ALLO$_2$

Bermúdez-Otero 2012, 2013; Paster 2014; Armstrong 2021; Bonet 2023

A. Automatic allomorphy

Stop aspiration in stressed syllable onset (Odden, 2005)

“Alternations involving aspiration” (Odden, 2005). We assume one underlying form with /p t k/, and a rule that derives the exponent with an aspirated [pʰ tʰ kʰ] when it is realised initially in an stressed syllable.

Coda, [+stress] σ		Onset, [-stress] σ		Onset [+stress] σ	
hɛl**p**	help	ˈhɛl.**p**-ə	helper	hɛl.ˈ**pʰ**-i	helpee
græn**t**	grant	ˈgræn.**t**-ə	grantor	græn.ˈ**tʰ**-i	grantee
straɪ**k**	strike	ˈstraɪ.**k**-ə	striker	straɪ.ˈ**kʰ**-i	strikee

How about automatic allomorphic in Mapudungun?

Surface allomorphy of the indicative marker:

Underspecified high front segment /I/ is realised as a high front vowel [i] after a consonant or [j] after a vowel (Molineaux 2014). Alternatively, /j/ is underlying, and [-i] derived by rule (Smeets 2008).

After a vowel		After a consonant	
a.**mu-j**	go-IND.3	ki.**m-i**	know-IND.3
kɨ.**pa-j**	come-IND.3	le.**f-i**	run-IND.3
ʈʂa.w**ɨ-j**	gather-IND.3	tu.**w-i**	be.from-IND.3

B1. Optimising suppletive allomorphy

Indefinite article in English as optimising

before **V**	before **C**
an=**a**pple	**a**=**b**ook
an=**e**gg	**a**=**t**ree

Why is optimising? (Mascaró 1996, *pace* Pak 2016)

***VV**	***CC**
***a**=**a**pple	***an**=**b**ook
***a**=**e**gg	***an**=**t**ree

"*a* is only *an* with his coat off"
(Nesbitt, 1885)

Mascaró 1996, Nesbitt 1885

How about optimising PCSA in Mapudungun?

Allomorphy of the collective suffix. Two possibilities for a mono-representational analysis (to be discarded):

1) Epenthesis of [e]? Nope, [ɨ] is the canonical epenthetic vowel (Smeets 2008).
2) Deletion of [e]? Homophonous *-entu* 'take away' does not change in the same way after a vowel-final verb: [wiʐ**u-e**ntu-n, nɨl**a-e**ntu-n], etc.

After a consonant		After a vowel	
θeɣɨ**ɲ-entu**	volcano-COLL	katʃ**u-ntu**	grass-COLL
mamɨ**ʎ-entu**	wood-COLL	kuʐ**a-ntu**	stone-COLL
wa**w-entu**	valley-COLL	mawiθ**a-ntu**	hill-COLL
kɨʎe**ɣ-entu**	hillside-COLL	ʐɨm**e-ntu**	reed-COLL

Why optimising?

***CC**
θeɣɨ**ɲ-ntu**
mamɨ**ʎ-ntu**

B2. Arbitrary suppletive allomorphy

Campan languages (Arawakan)

Ashé-Ashá languages belong to the Campan subgroup. Ashéninca is an Ashé-Ashá language and can divided into five regional variations:

- Pajonal,
- Pichis,
- Ucayali,
- Perené
- Apuracayali.

(Pedrós, 2018)

Rough location of Campan languages based on Pedrós 2018, done using Google Earth

Arbitrary PCA of the possession suffix in some Campan languages

After disyllabic or bimoraic stems, ***-ni/-ne*** is used, while after stems with more syllables of morae, ***-ti/-te*** is used: Apurucayali Ashéninca o Axininca (Payne 1981:101), Nanti (Michael 2008:300), Alto Perené (Mihas 2015:335, 2010) and Ucayali-Pajonal (Pedrós 2023:179).

	After bimoraic stems		After more than two morae	
Axininca	no-**mii-ni**	1p-otter-poss	no-**maini-ti**	1-bear-POSS
Nanti	i-**gusi-ne**	3p.m-guan-poss	o-**chaberi-te**	3.F-chicken-POSS
Alto Perené	i=**mapi-ni**	3.M=stone-poss	n=**otzitsi-te**	1=dog-POSS
Ucayali-Pajonal	i-**koka-ni**	3p.m-coca-poss	i-**koshina-ti**	3.M-kitchen-POSS

Are there arbitrary PCSA phenomena in Mapudungun?

Kɨ-allomorphy is PCSA

pi-le-	'to be saying'	aθ-kɨle-	'to in order'
aʎfɨ-le-	'to be injured'	ɲikɨf-kɨle-	'to be silent'
loŋko-le-	'to be acting as a leader'	pot͡ʂoŋ-kɨle-	'to be bent down'
afmatu-le-	'to be surprised'	ʐakiθuam-kɨle-	'to be thinking'

pi-jaw-	'to go around saying'	lef-kɨjaw-	'to be running around'
ɨta-jaw-	'to be grazing around'	kɨθaw-kɨjaw-	'to work here and there'
kisu-jaw-	'to be around on one's own'	ʐakiθuam-kɨjaw-	'to go around thinking'
t͡ʂeka-jaw-	'to walk about'	wakeɲ-kɨjaw-	'to go around making noise'

It is not automatic! No phonological process deletes [kɨ] from morphemes. [kɨ] appears in native roots: /kɨl̪en̪/ 'tail', /kɨla/ 'three', /ʐekɨl-/ 'to lean against'), affixes (/-kɨnu/ progressive; /-kɨlo/ (archaic) 'help to Verb'), as well as loanwords: /kɨlafu/ 'nail' < Spanish *clavo*, /sikɨleta/ 'bycicle' < Spa. *bicicleta*

But is it arbitrary or optimising?

Kɨ-allomorphy is input-conditioned (Paster, 2015)

[w]-vocalisation is predicted from an underlying /w/ in the stem (Salas 2006).

/epew/ → [epew] ~ [epeo] '*epew*, a traditional Mapuche story'

Affix [-kɨle] appears after a stem ending in a consonant —the underlying final segment—, regardless of the surface realisation of the word with a vowel.

/epew-kɨle-tu-j/ → ✅ [epe**o-**kɨle-tu-j]

❌ [epeo-le-j]}

'she is telling an *epew*'

(Salas 2006:202)

Apparent misapplication due to later phonological processes

Morphological selection		/apo/+/le/+/j/ full-STA-IND.3	/epew/+/kɨle/+/j/ story-STA-IND.3
		↓	↓
Underlying representations		/apo-le-j/	/epew-kɨle-j/
Stem level	*Syllabification*	a.po	e.pew
Word level	*Resyllabification*	a.po.lej	e.pew.kɨ.lej
Phrase level	*/w/-vocalisation*	—	e.pe**o**.kɨ.lej
Surface forms		[a.po.lej]	[e.peo.kɨ.lej]

Not-optimising (Paster 2015)

- In monosyllabic stems, selection of [-kɨle] could be analysed as an optimal repair to avoid adjacent stresses, a dispreferred configuration (Molineaux 2023)
  - Multi-suffix verbs in Mapudungun have two stresses (Molineaux 2018, 2023)
  - [-(kɨ)lé] and [-(kɨ)jáw] can attract stress on the final syllable

Predicted derivation (under a disjunctive input analysis):

❌ /pí-**{lé, kɨlé}**-n/ → *[pí-**kɨ**lé-n] ‘I am saying’

However, this prediction is not fulfilled:

✅ /pí-lé-n/ → Clash resolution → [pi-lé-n] ‘I am saying’

Modelling interaction between morphology and phonology

Anatomy of a Lexical entry (a lá Jackendoff 1997)

Attributes in SEM, SYN, PHON

Informal notation for semantics

Phonological form

$$
\begin{bmatrix}
\textbf{SEM:} & \text{‘go’} \\
\textbf{PHON:} & /\text{amu}/ \\
\textbf{SYN:} & [_{\mathrm{V}}[_{\mathrm{V}}\ \underline{\quad\quad}\ ][_{\mathrm{Aff}}\ \cdots\ ]]
\end{bmatrix}
$$

Syn is represented with bracketed hierarchy and contains the morphosyntactic categories.

The symbol “...” represents a subcategorisation frame.

Jackendoff 1997, Ackema and Neeleman 2007, Bermudez Otero 2012

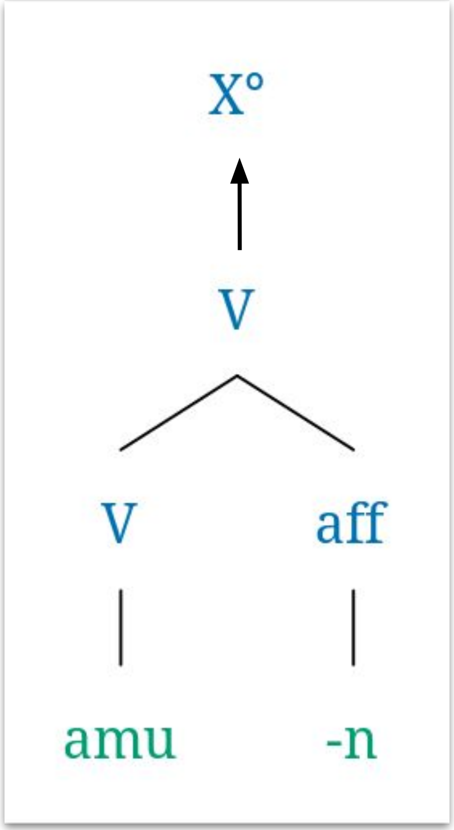
Lexical entries with morphological subcategorisation

SEM: ‘go’	**SEM:** ‘IND.1.SG’
PHON: /amu/	**PHON:** /-n/
SYN: $[_V [_V \_\_ [_{Aff} \cdots]]]$	**SYN:** $[_V [_V \cdots [_{Aff} \_\_]]]$

The insertion of the items is done considering the subcategorisation frames:

An affix like /-n/ IND.1SG subcategorises for a verbal base:

$$[_V [_V \text{amu}] [_{aff} \text{-n}]] >> /\text{amun}/ \text{ ‘I go’}$$



Subcategorisation can carry phonological information

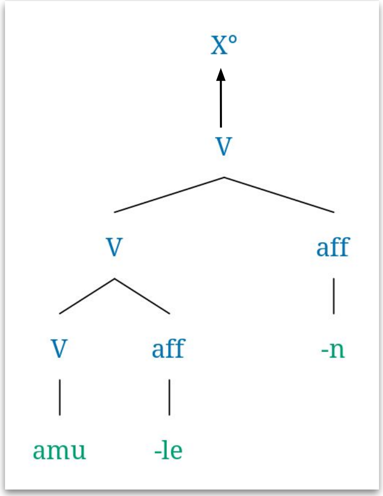
SEM: ‘go’
PHON: /amu/
SYN: $[_V [_V \_\_ \ [_{Aff} \dots]]$

SEM: ‘run’
PHON: /lef/
SYN: $[_V [_V \_\_ \ [_{Aff} \dots]]]$

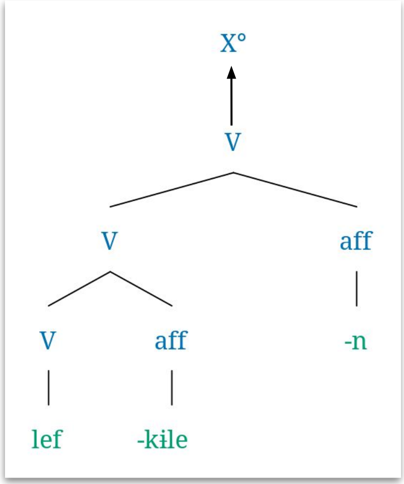
SEM: STA / PROG
PHON: /le/ V
SYN: $[_V \dots | [_{Aff} \_\_] \dots]$

SEM: STA / PROG
PHON: /kɨle/ C
SYN: $[_V \dots | [_{Aff} \_\_] \dots]$

SEM: ‘IND.1.SG’
PHON: /-n/
SYN: $[_V [_V \dots [_{Aff} \_\_]]]$



[amu-le-n]
go-STA-IND.1sg
‘I am going’



[lef-kɨle-n]
run-STA-IND.1sg
‘I am running’

Conclusions

Conclusions

- Kɨ-allomorphy is phonologically-conditioned suppletive allomorphy (PCSA), and despite similarities, it is not phonologically optimising, but input subcategorising.
- In a language with complex morphology, we see the same patterns that we see cross-linguistically.
- The “missed generalisation” can be explained by other paths, such as diachrony. This is my next step.
- Less-studied languages can help us elucidate the division of labour between storage and computation, a fundamental question in a modular conceptualisation of the grammar architecture.

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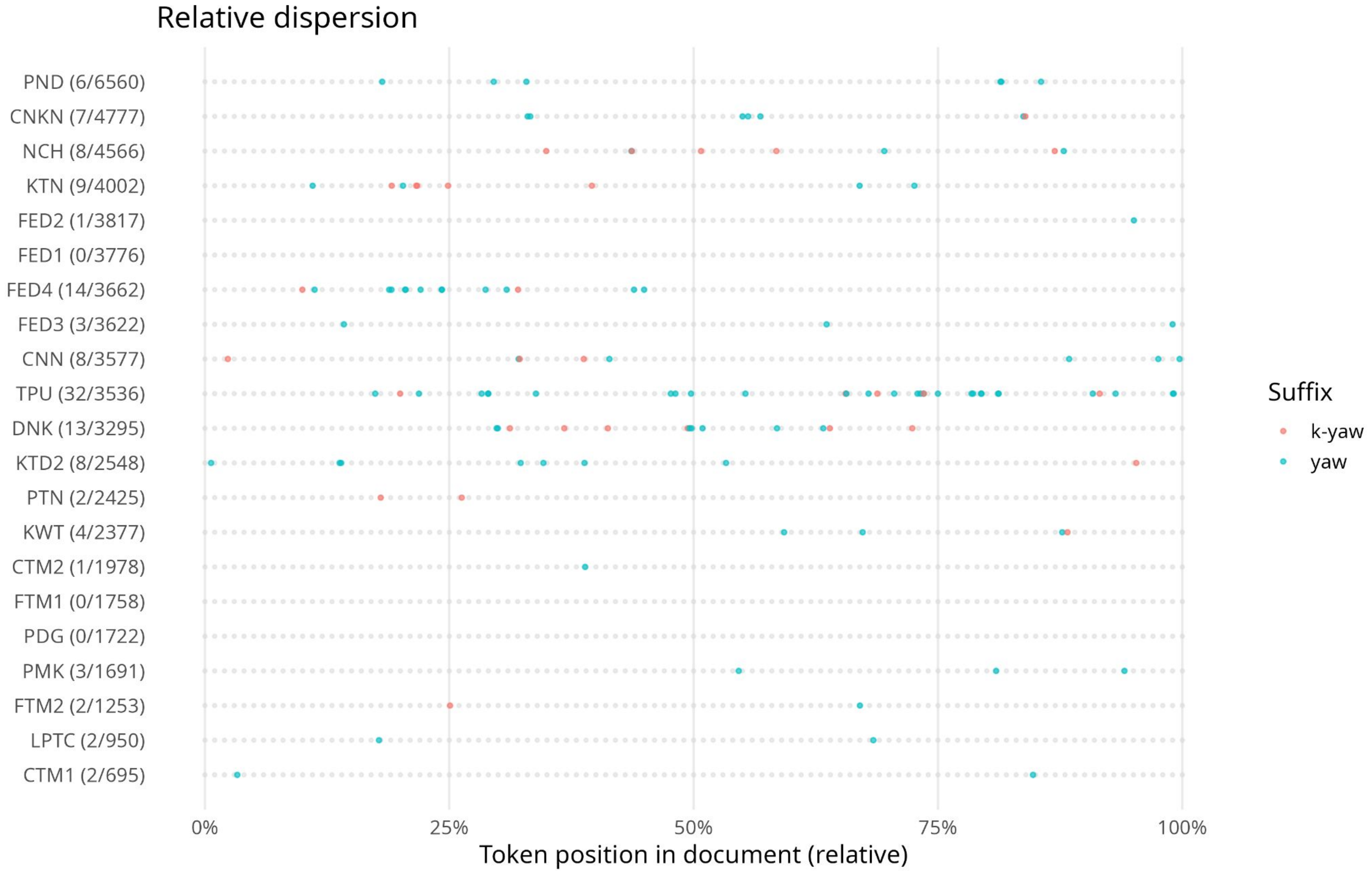
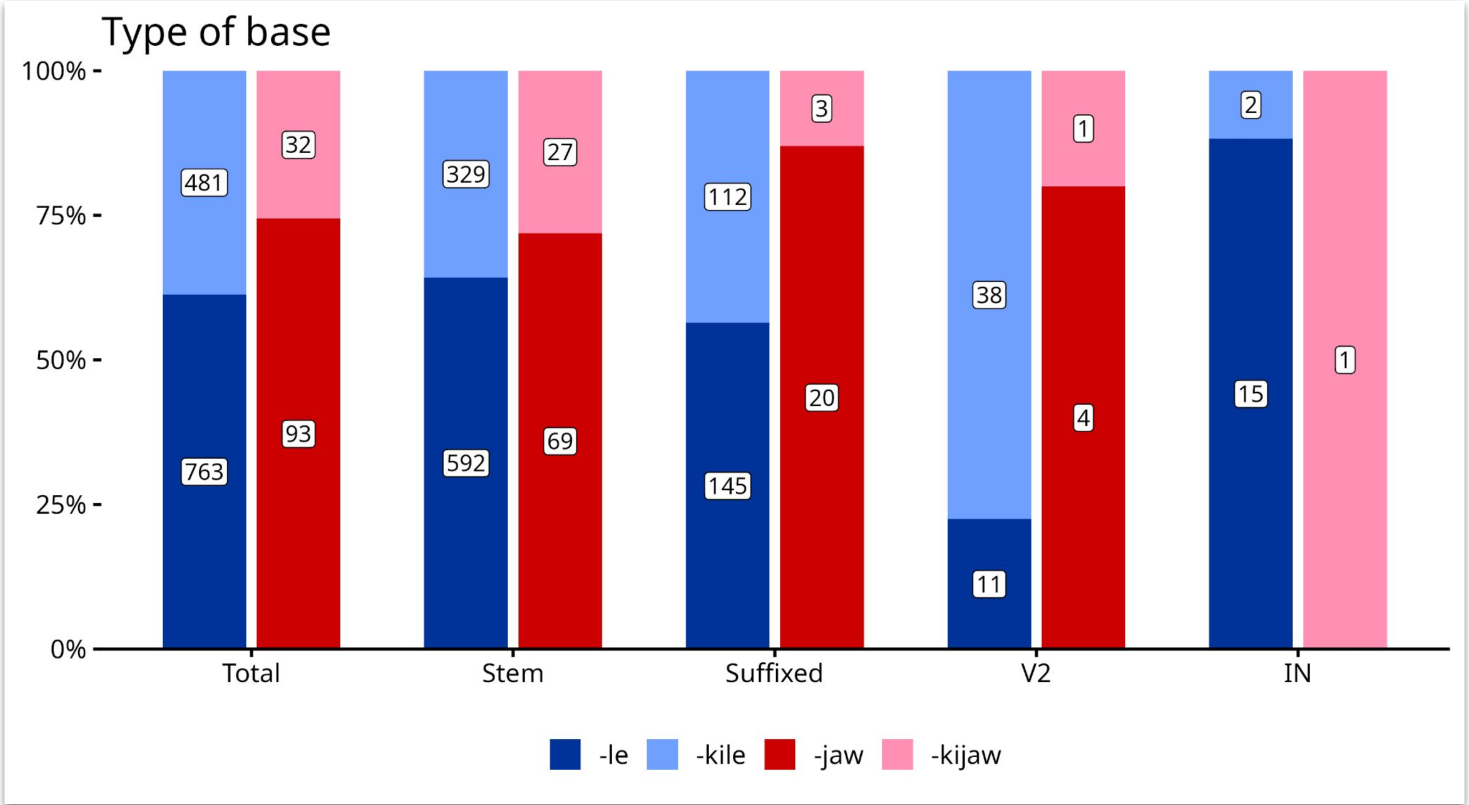
Appendices

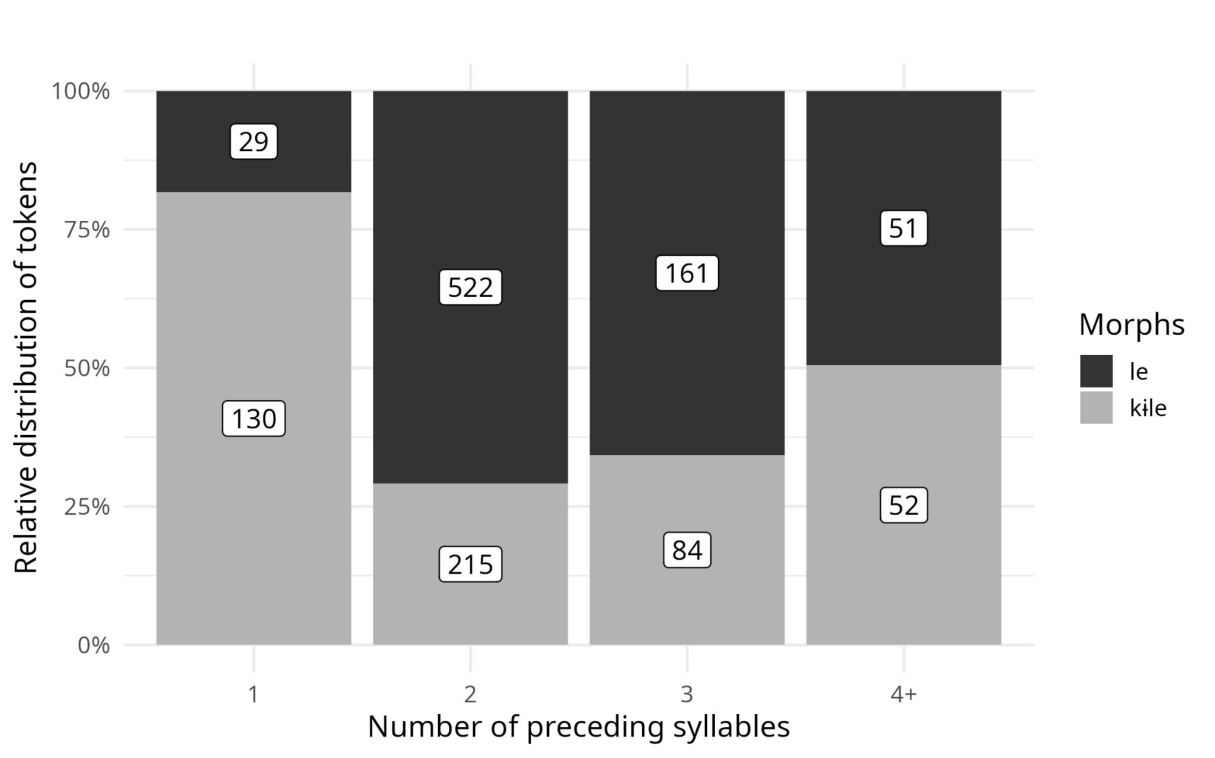


Relative dispersion
PND (6/6560)
CNKN (7/4777)
NCH (8/4566)
KTN (9/4002)
FED2 (1/3817)
FED1 (0/3776)
FED4 (14/3662)
FED3 (3/3622)
CNN (8/3577)
TPU (32/3536)
DNK (13/3295)
KTD2 (8/2548)
PTN (2/2425)
KWT (4/2377)
CTM2 (1/1978)
FTM1 (0/1758)
PDG (0/1722)
PMK (3/1691)
FTM2 (2/1253)
LPTC (2/950)
CTM1 (2/695)
0%
25%
50%
75%
100%
Token position in document (relative)
Suffix
k-yaw
yaw

Results considering the type of base



Results considering the number of syllables in the base



B. Suppletive allomorphy that is optimising

Six classes of optimising phonological conditions can force allomorph choice in Suppletive PCA (Nevins 2011):

1. Segmental dissimilation
2. Segmental phonotactics
3. Syllable structure
4. Morphological alignment
5. Stressedness and vowel quality
6. Foot structure

Segment-level phenomena (*pace* Scheer, 2023)

Syllable-level phenomena

Prosodic-level phenomena

"In short, allomorphy never produces ill-formed syllables" (Kager 1996:155)

Why PCA phenomena are important?

Where is the separation between storage and computation?

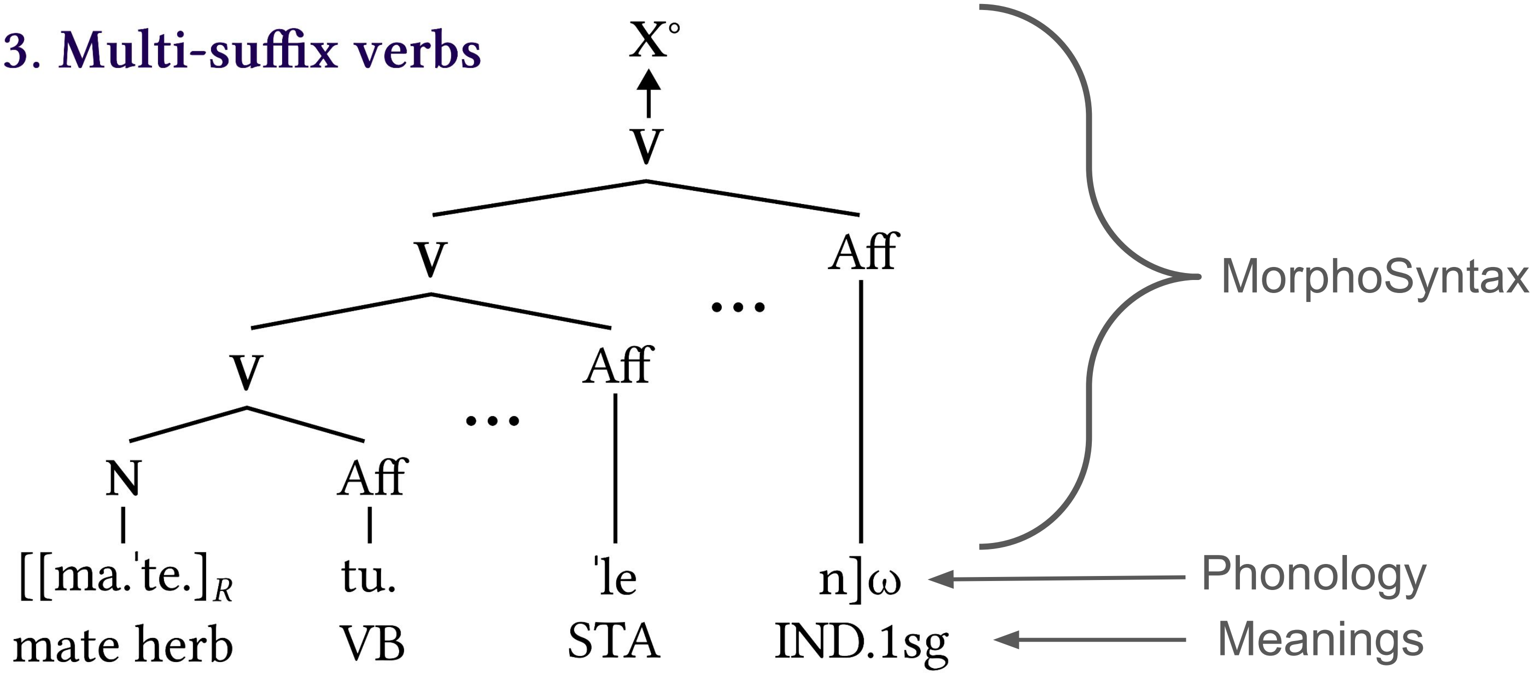
"where in the grammar are lexical, impredictable morpheme alternations to be included, and where are phonological regularities to be expressed" (Mascaró, 1996:473)

"Are morphophonological alternations the result of phonological rules, or do they result from the storage in memory of distinct allomorphs?" Embick and Shwayder's FQM (2018:193)

"The most active questions of debate thus revolve around whether, given that some **allomorph selection is phonologically optimizing**, this should be built into the architecture of the grammar, or whether on the contrary, given that some allomorph selection is not phonologically optimizing, a single mechanism that simply lists phonological environments is all that is needed." (Nevins 2011)

Assumed morphophonology of complex words in Map.

2.3. Multi-suffix verbs



‘I am drinking mate (a herb).’

Molineaux 2023, Bickel & Zúñiga 2017

Expressing modularity: strict separation of different representations